

NAUFAL ICHSAN

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SUMMARY

Recent Computer Science graduate from Universitas Indonesia with a strong foundation in designing and scaling end to end software systems across full stack frameworks, enterprise data pipelines, and cloud native infrastructure. Demonstrates deep technical versatility in Python, Java, Go, and Scala with a proven track record of engineering resilient event driven architectures, optimizing large scale distributed data processing, automating cloud deployments, and integrating AI capabilities directly into high performance software solutions. A highly adaptable problem solver and strong communicator who applies core system design principles to collaborate across cross functional teams.

EDUCATION

Universitas Indonesia

Depok, West Java

Bachelor of Computer Science, Faculty of Computer Science, cGPA: 3.73/4.00

2022 - 2026

- Relevant Coursework: Software Engineering, Computer Network, Platform Based Programming, Data Structure & Algorithm, Software Architecture, Software Security Design, Operating System, Data Science & AI, Cloud Computing.

EXPERIENCE

Accenture

Jakarta, Indonesia

Technology Consulting Intern

March 2026 - Current

- Engineered 10+ event-driven backend services using Temporal, improving fault tolerance and production reliability across distributed systems.
- Built dynamic Kafka and Temporal test consumers, reducing system integration testing setup effort by 60% and accelerating workflow validation.
- Automated kubernetes deployments by migrating Jenkins pipelines to GitLab CI/CD, reducing deployment time by 70% and minimizing manual configuration errors.
- Collaborated with 30+ engineers, system architects, QA engineers, and client stakeholders to translate business requirements into scalable backend solutions across six agile sprints.

Traveloka

Jakarta, Indonesia

Software Engineer Intern - Data Platform

September 2025 - March 2026

- Engineered end-to-end ETL pipelines using Apache Spark and Scala, processing large-scale financial datasets and improving pipeline performance by 30%.
- Built event-driven data pipelines using AWS Lambda, replacing legacy schedulers and reducing data latency and pipeline failures by 50%.
- Automated AWS infrastructure provisioning using Terraform (IaC), standardizing environments across development, testing, and production.
- Developed data validation and unit testing frameworks for millions of financial records, maintaining 100% pipeline test coverage.
- Built developer automation tools using Python and VB.NET for OneStream metadata analysis, reducing manual impact analysis time by 75%.

PT Pejuang Indonesia Cerdas

Jakarta, Indonesia

Software Engineer Intern

January 2025 - May 2025

- Led the software system architecture design and selected the technology stack for a scalable online quiz platform, defining the backend, infrastructure, and deployment architecture.
- Developed RESTful APIs using Django, leveraging Redis caching and Celery asynchronous processing to support 1,000+ concurrent users on limited infrastructure while improving answer submission performance by 60%.
- Containerized applications using Docker Compose and automated deployments with GitHub Actions CI/CD, reducing environment setup time by 80% while securing production with Nginx, HTTPS (Certbot), and API rate limiting.
- Implemented unit and integration testing with 90%+ backend logic coverage, established Prometheus and Grafana monitoring, and maintained 100% deployment uptime through blue-green deployments.

- Mentored 50+ students in different core CS topics (OOP, DSA, Computer Network, Web Programming), translating complex technical concepts into clear, accessible explanations for students with varying levels of technical knowledge.
- Designed 30+ lab sessions and facilitated real-time troubleshooting and architectural guidance, helping students understand system design decisions and reducing common implementation errors by 40%.
- Simplified technical concepts and design trade-offs, enabling students to quickly evaluate and implement effective solutions.

LEADERSHIP & ACTIVITY

- Strengthened BEM Fasilkom UI's branding by designing and producing visual content and official videos across social media, websites, and campaigns.
- Boosted audience engagement and brand recognition through consistent, high-impact media and creative content strategy.
- Collaborated with cross-functional teams to ensure cohesive communication and unified branding across departments.

PROJECT

FastAPI (Python), Next.js (TypeScript), LangGraph, PostgreSQL, Docker, Git

- Architected an end-to-end AI resume tailoring platform utilizing a Next.js (TypeScript) frontend and an asynchronous FastAPI (Python) backend, serving optimized PDFs to users.
- Designed async REST APIs with FastAPI, SQLAlchemy, and asyncpg, modeling PostgreSQL data for user profiles, OAuth accounts, and generation history.
- Hardened API security with JWT-based authorization, SlowAPI rate limiting, and PDF validation to mitigate malicious file/path traversal attacks.

Go, Kubernetes, Docker, AWS, Prometheus, Cilium (eBPF), Git

- Designed and implemented a network-aware Kubernetes Descheduler PreEviction plugin to incorporate node-to-node network latency into workload rescheduling decisions.
- Built the plugin in Go, integrating Cilium and Prometheus for network telemetry and latency-aware eviction.
- Conducted performance testing on AWS EC2 and documented results, demonstrating a 30–50% reduction in cluster network latency compared with the default descheduling behavior without the custom plugin.

TECHNICAL SKILLS

Languages: Python, Java, Scala, Kotlin, Go, JavaScript, TypeScript
Frameworks & Libraries: Spring Boot, Django, Next.js, Spark Scala, Gin
Databases & Caching: PostgreSQL, MongoDB, Redis
Distributed Systems & Messaging: Kafka, RabbitMQ
DevOps & Orchestration: Git, Docker, Kubernetes, GitHub/Gitlab Actions, Jenkins, Temporal
Monitoring & Observability: Prometheus, Grafana
Cloud & Infrastructure: Cloud-native, AWS, Google Cloud (GCP), Nginx
Development Practices: Agile (Scrum), TDD, Microservices, Clean Code, Design Pattern, Event-Driven

LANGUAGES & INTEREST

Languages: Indonesian (Native), English (Professional Working), Japanese (Limited Working)
Interests: Software Development, Cloud Infrastructure & DevOps, System Design, AI Engineering